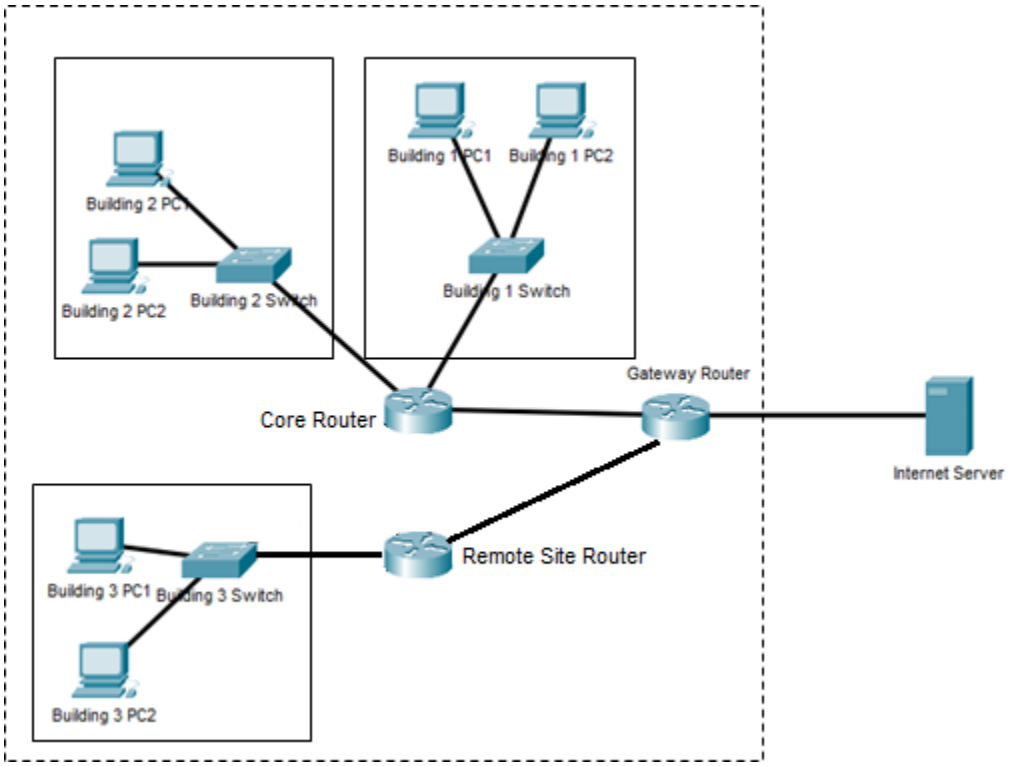
Design Project – INFR1411U

Instructions:

This is a group assignment. You will work on this assignment with your lab partner, if you have one. Each member of a lab team will submit **the same copy of the design assignment** through Canvas.

Design project)

The following diagram shows the topology of a corporate network consisting of three buildings. Each building contains a number of PCs and a switched LAN.



Two buildings are connected to a core router, which is connected to a Gateway router. A remote site (Building 3) is connected to a remote site router that is connected separately to the Gateway router. The Gateway router has a single connection to an Internet server with IP address **108.0.0.1**. The link between the Gateway router and the Internet server uses the subnet **108.0.0.0/30**. You can find the network map in the following:

The following list include IP address space requirement for each building (hosts and network devices), excluding the routers:

Building 1	132 IP addresses
Building 2	250 IP addresses

Building 3	254 IP addresses

VLSM is implemented. In order to determine the base network address for each building, use your and your partner's student ID numbers for the buildings in the following way:

1. Take your nine-digit student ID 100xxyyzz, and create an IP address in form of 100.xx.yy.zz from it, and calculate the network address associated with this IP address and the corresponding subnet mask for Building 1 for accommodating the hosts in Building 1.
2. Your lab partner should do the same for building 2 (decide between yourselves who will design which building).
3. For building 3, assign all addresses from the **192.168.0.0** range with the appropriate subnet mask.
4. Choose private IP addresses of your choice for the connection between the routers.

If you have no lab partner and work individually, create a network address from your own student ID based on the above instructions, and subnet it for all subnets within the corporate network.

Assignment

a. Based on the above requirements, design an IPv4 addressing scheme for all subnets in this network. Your scheme should include the network address and subnet mask for each subnet. The subnet addresses should not have any overlaps.
(25 Marks)

building 1

- -132 required hosts
- -network address 100.98.0.0/24
- -subnet mask 255.255.255.0
- Default gateway (Core router G0/1): 100.98.0.1

Building 2

- - required hosts 250
- -network address 100.98.1.0/24
- -subnet mask 255.255.255.0
- Default gateway (Core router G0/0): 100.98.1.1

Building 3

- -required hosts 254 hosts
- -network address 192.1680.0/24
- -subnet mask 255.255.255.0
- Default gateway (Remote router G0/0): 192.168.0.1

Gateway to the Internet server

- -host 2
- -network address 108.0.0.0/30
- -subnet mask 255.255.255.252
- Gateway router (Gig0/1): 108.0.0.2
- Internet server: 108.0.0.1

Router 1-3

- 2 hosts
- 10.1.1.0/30
- Subnet mask 255.255.255.252
- Core router S0/1/0: 10.1.1.1
- Gateway router S0/1/0: 10.1.1.2

Router 2-3

- 2 hosts
- Network address 10.1.1.4/30
- Subnet mask 255.255.255.252
- Gateway router S0/1/1: 10.1.1.5
- Remote router S0/1/0: 10.1.1.6

b. Determine and show the static routes required at each router to allow successful pings between PCs in different buildings, and from each PC to the Internet server. Do NOT deploy any dynamic routing protocol. Also, show the content of the routing table at each router
(25 Marks)

Core router

ip route 192.168.0.0 255.255.255.0 10.1.1.2

ip route 108.0.0.0 255.255.255.252 10.1.1.2

```
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.1/32 is directly connected, Serial0/1/0
     100.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       100.98.0.0/24 is directly connected, GigabitEthernet0/1
L       100.98.0.1/32 is directly connected, GigabitEthernet0/1
C       100.98.1.0/24 is directly connected, GigabitEthernet0/0
L       100.98.1.1/32 is directly connected, GigabitEthernet0/0
     108.0.0.0/30 is subnetted, 1 subnets
S       108.0.0.0/30 [1/0] via 10.1.1.2
S    192.168.0.0/24 [1/0] via 10.1.1.2

Router#
```

Gateway router

ip route 100.98.0.0 255.255.255.0 10.1.1.1

ip route 100.98.1.0 255.255.255.0 10.1.1.1

ip route 192.168.0.0 255.255.255.0 10.1.1.5

```
Router>
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.2/32 is directly connected, Serial0/1/0
C       10.1.1.4/30 is directly connected, Serial0/1/1
L       10.1.1.6/32 is directly connected, Serial0/1/1
     100.0.0.0/24 is subnetted, 2 subnets
S       100.98.0.0/24 [1/0] via 10.1.1.1
S       100.98.1.0/24 [1/0] via 10.1.1.1
     108.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       108.0.0.0/30 is directly connected, GigabitEthernet0/1
L       108.0.0.2/32 is directly connected, GigabitEthernet0/1
S    192.168.0.0/24 [1/0] via 10.1.1.5
 --More--
```

Remote site router
ip route 100.98.0.0 255.255.255.0 10.1.1.6
ip route 100.98.1.0 255.255.255.0 10.1.1.6
ip route 108.0.0.0 255.255.255.252 10.1.1.6

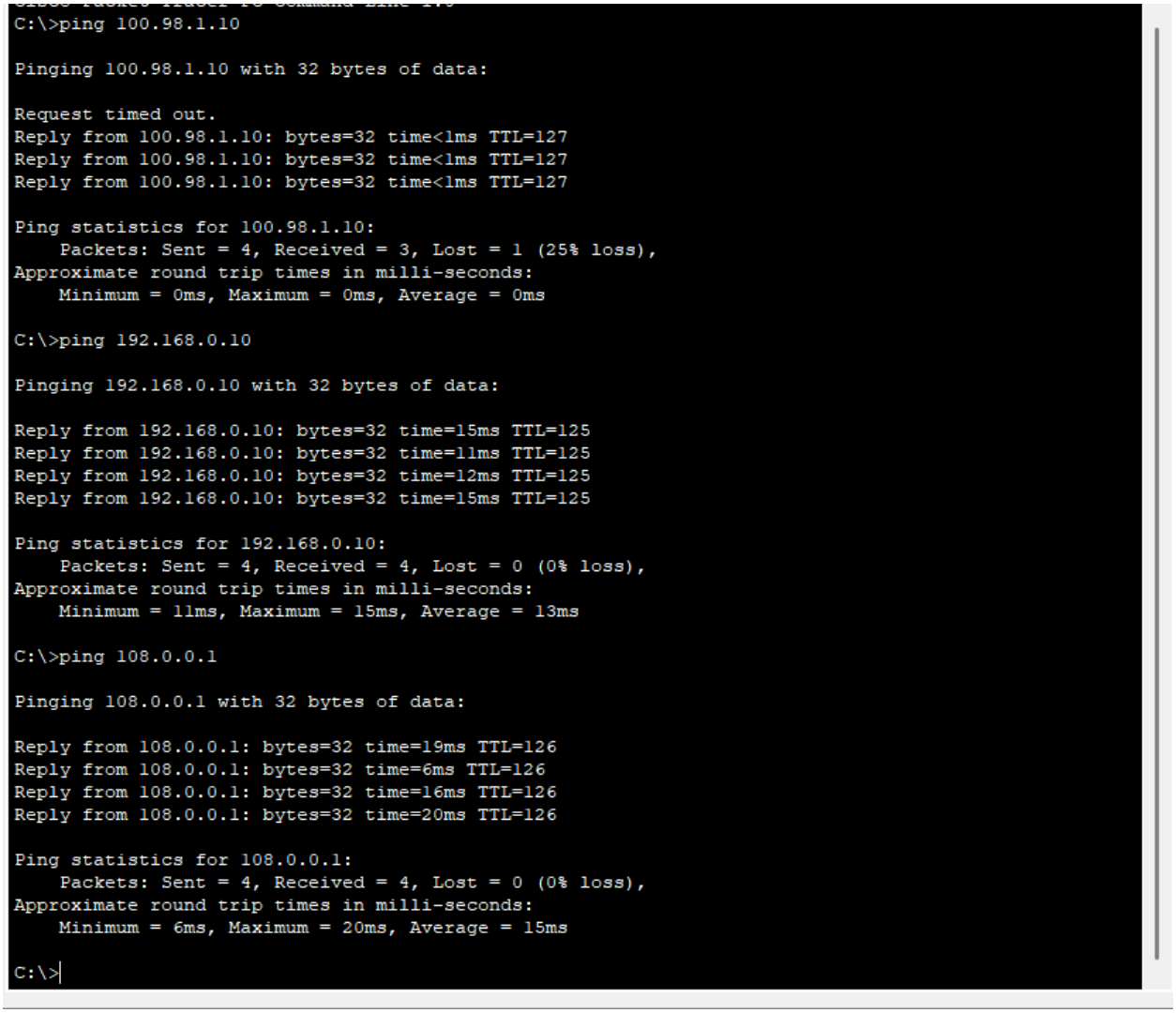
```
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.4/30 is directly connected, Serial0/1/0
L       10.1.1.5/32 is directly connected, Serial0/1/0
     100.0.0.0/24 is subnetted, 2 subnets
S       100.98.0.0/24 [1/0] via 10.1.1.6
S       100.98.1.0/24 [1/0] via 10.1.1.6
     108.0.0.0/30 is subnetted, 1 subnets
S       108.0.0.0/30 [1/0] via 10.1.1.6
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0

Router#
```

c. Implement this network in Packet Tracer using two hosts for each building. Use your addressing design from (a) and static routes from (b). Confirm that the hosts in different buildings can ping each other and the Internet server. Ask your lab instructor to verify and confirm the pings.



The pings were used to test connectivity between the internet server and all subnets, and the test was successful.

(40 Marks)

d. Submit a feedback and reflection report (separate file)
(10 marks)

Deliverables

Each team member must submit a zip file containing the following through Canvas:

- An MS-word file containing your answers to (a) and (b).
- The packet Tracer file for (c). You will receive zero mark for part (c) if the packet tracer file is not attached, even if your instructor confirms the work.

- A completed feedback and reflection report.

(End)